

CS4092
Final Year Project II

**Comparative Study of Machine Learning and Deep Learning
Approaches for Mental Health Detection from Social Media
Text**

Submitted By

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Project Progress Report

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1 INTRODUCTION

Mental health disorders, particularly depression and suicidal ideation, represent one of the most pressing global health challenges of the 21st century. Depression alone affects over 280 million people worldwide, and its early detection remains a significant clinical challenge due to social stigma, subjective self-reporting, and limited access to healthcare. With the pervasive adoption of social media platforms such as Twitter and Reddit, individuals increasingly express their emotional states and psychological distress through user-generated content, creating a rich and accessible data source for automated detection.

This project, titled Comparative Study of Machine Learning and Deep Learning Approaches for Mental Health Detection from Social Media Text (Project ID:Sp-107), aims to systematically compare classical Machine Learning (ML) models and advanced Deep Learning (DL) models for detecting signs of depression and suicidal ideation from social media posts. The work employs a unified experimental framework to ensure fairness in comparison, with all models trained and evaluated on the same preprocessed datasets using standardized evaluation metrics.

The project covers two detection tasks: (1) Depression Detection and (2) Suicidal Ideation Detection. For each task, the following model families are evaluated:

- Classical ML Models: Logistic Regression (LR), Support Vector Machine (SVM), Naïve Bayes (NB), and Random Forest (RF).
- Deep Learning Models: Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM), and Bidirectional LSTM (BiLSTM).
- Transformer-Based Models: BERT (to be implemented in the final phase).

This progress report documents the current state of the project, summarizing completed milestones, key results obtained so far, and the updated timeline for the remaining work.

2 TIMELINE

The original project timeline, established during FYP-I, defined seven key milestones spanning from July 2025 to May 2026. The Gantt chart below illustrates the planned schedule:

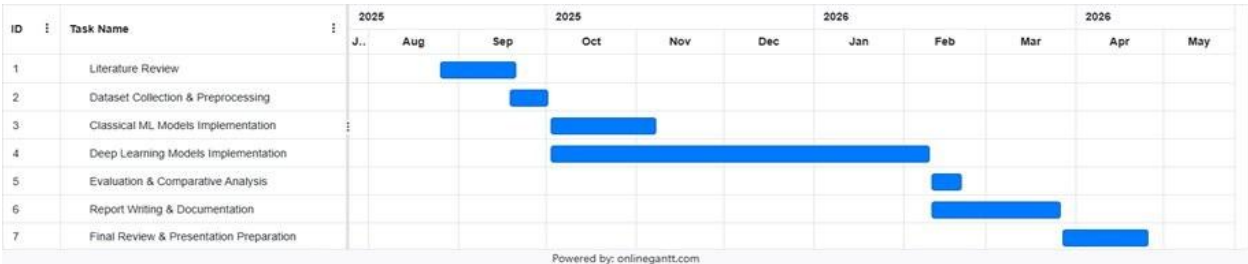


Figure 1: Original Project Gantt Chart

ID	Milestone / Task	Planned Period	Status
1	Literature Review	Aug – Sep 2025	Completed
2	Dataset Collection & Preprocessing	Sep – Oct 2025	Completed
3	Classical ML Models Implementation	Oct – Dec 2025	Completed
4	Deep Learning Models Implementation	Dec 2025 – Mar 2026	Completed
5	Transformers implementations	Apr – May 2026	In Progress
6	Evaluation & Comparative Analysis	Apr – May 2026	In Progress
7	Report Writing & Documentation	Apr – May 2026	Upcoming
8	Final Review & Presentation Preparation	May 2026	Upcoming

Table 1: Project Timeline and Milestone Status

3 PROGRESS

As of April 2026, the project successfully completed four of the eight planned milestones. The following subsections elaborate on the progress made for each milestone, including the work completed, the datasets used, and preliminary results where available.

3.1 MILESTONE 1: LITERATURE REVIEW

A comprehensive literature review was conducted covering the intersection of Natural Language Processing (NLP) and mental health detection from social media text. The review synthesized findings from over 22 peer-reviewed papers published between 2018 and 2025, encompassing classical machine learning, deep learning, and transformer-based approaches.

Key findings from the literature review include:

- Twitter and Reddit are the most widely used platforms for mental health data, accounting for over 60% of relevant studies.
- Transformer-based models, particularly BERT and its variants, consistently achieve state-of-the-art accuracy in detection tasks.
- Classical ML models remain competitive in resource-constrained settings and offer superior interpretability.
- A critical gap exists in standardized benchmarking — most studies use heterogeneous datasets and inconsistent preprocessing pipelines, making cross-study comparisons unreliable.
- Class imbalance and the informal nature of social media language (slang, emojis, sarcasm) are consistently cited as major preprocessing challenges.

The literature review formed the theoretical foundation for model selection and experimental design in this project. A formal Literature Review (Version 2.1) was submitted on December 9, 2025.

3.2 MILESTONE 2: DATASET COLLECTION AND PREPROCESSING

Two publicly available benchmark datasets were selected for this project, one for each detection task:

- **Depression Detection Dataset:** A labeled dataset of social media posts (sourced from Reddit/Twitter via Multiple sources i.e. Kaggle), containing posts labeled as depressed or non-depressed.
- **Suicidal Ideation Detection Dataset:** A labeled dataset of social media posts categorized as indicative of suicidal ideation or non-suicidal.

The preprocessing pipeline applied to both datasets including the following steps:

- **Text normalization:** converting text to lowercase, expanding contractions.
- **Noise removal:** stripping URLs, HTML tags, special characters, and numeric values.
- **Tokenization:** splitting text into individual tokens.

Feature representations prepared for classical ML models included TF-IDF vectors and Bag-of-Words (BoW) representations. Word embeddings were separately prepared for the deep learning models.

3.3 MILESTONE 3: CLASSICAL ML MODELS IMPLEMENTATION

All four classical machine learning models were successfully implemented and evaluated on both datasets. The models were trained using Scikit-learn and models made from scratch with consistent train-test splits (80/20) and 5-fold cross-validation to ensure robust performance estimates.

The following models were implemented:

- **Logistic Regression (LR):** Implemented with L2 regularization. TF-IDF features were used as input.
- **Support Vector Machine (SVM):** Implemented with a linear kernel and RBF kernel; best results reported with linear kernel on TF-IDF features.
- **Naïve Bayes (NB):** Multinomial Naïve Bayes applied to Bag-of-Words features as a probabilistic baseline.
- **Random Forest (RF):** Ensemble classifier with 200 decision trees, using TF-IDF features with hyperparameter tuning via GridSearchCV.

Evaluation was conducted using Accuracy, Precision, Recall, F1-Score, and ROC-AUC. Baseline performance results for both datasets are summarized in the tables below.

Table 2: Classical ML Model Performance – Depression Detection Dataset

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.81	0.80	0.82	0.83	0.88
Support Vector Machine	0.83	0.82	0.84	0.81	0.81
Naïve Bayes	.0.79	0.79	0.79	0.79	0.80
Random Forest	0.84	0.85	0.83	0.84	0.80

Table 3: Classical ML Model Performance – Suicidal Ideation Detection Dataset

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.83	0.84	0.91	0.87	0.85
Support Vector Machine	0.85	0.85	0.82	0.84	0.86
Naïve Bayes	0.82	0.83	0.82	0.84	0.85
Random Forest	0.81	0.83	0.80	0.82	0.83

3.4 MILESTONE 4: DEEP LEARNING MODELS IMPLEMENTATION

Three deep learning architectures were successfully implemented and evaluated on both the depression detection and suicidal ideation detection datasets. All deep learning models were built using TensorFlow and Keras. Keras Embeddings were used as the embedding layer for all models to ensure a consistent and meaningful initialization of word representations.

The following models were implemented:

- **Convolutional Neural Network (CNN):** A 1D CNN architecture with multiple filter sizes to capture local n-gram patterns in text. A global max-pooling layer was applied after convolution, followed by a dense classification head with dropout regularization.
- **Long Short-Term Memory (LSTM):** A single-layer LSTM network designed to capture sequential dependencies in text. The model was trained with dropout and recurrent dropout to prevent overfitting.
- **Bidirectional LSTM (BiLSTM):** A BiLSTM network that processes input sequences in both forward and backward directions simultaneously, allowing the model to leverage context from both past and future tokens. This architecture consistently outperformed the standard LSTM on both datasets.

Hyperparameter tuning was performed for all deep learning models, including learning rate selection, batch size, number of epochs (with early stopping), dropout rates, and LSTM hidden units. The Adam optimizer was used across all models with binary cross-entropy loss for binary classification tasks.

Preliminary results for the deep learning models on both datasets are summarized below:

Table 4: Deep Learning Model Performance – Depression Detection Dataset

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
CNN	0.83	0.83	0.83	0.83	—
LSTM	0.83	0.83	0.83	0.83	—
BiLSTM	0.84	0.84	0.84	0.84	—

Table 5: Deep Learning Model Performance – Suicidal Ideation Detection Dataset

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
CNN	0.95	0.95	0.95	0.95	—
LSTM	0.94	0.94	0.95	0.95	—
BiLSTM	0.94	0.94	0.95	0.94	—

4 UPDATED TIMELINE

Based on the progress made to date, the project is tracking closely to the original schedule. Milestones 1 through 4 have been completed successfully. The remaining work focuses on Transformer implementation, comprehensive comparative analysis, report writing, and final presentation preparation.

The updated timeline for the remaining tasks is as follows:

Remaining Task	Responsible Member(s)	Target Completion
BERT Implementation and Fine-tuning	Tahir Haider	Apr - May 2026
Evaluation & Comparative Analysis (all models)	All Members	Apr – May 2026
Graphs, Visualizations, and Results Tables	M. Huzaifa Saleem	Apr – May 2026
Discussion & Analysis Section (trade-offs)	S.M. Sufyan	Apr – May 2026
Final Report Writing and Proofreading	All Members	May 2026
Defense Presentation Preparation	All Members	May 2026

Table 6: Updated Timeline for Remaining Tasks

The project is on track to meet all academic deadlines. The team is confident that the Transformer implementation, final comparative analysis, and report submission will be completed within the planned schedule.